



CodeForge AI - Quick Start Guide

Prerequisites Checklist

Before you begin, ensure you have:

- [] **Python 3.11+** installed ([Download](https://python.org) (https://python.org))
 - [] **Node.js 18+** installed ([Download](https://nodejs.org) (https://nodejs.org))
 - [] **PostgreSQL 14+** installed ([Download](https://postgresql.org) (https://postgresql.org))
 - [] **Venice AI API Key** ([Sign up](https://venice.ai) (https://venice.ai))
 - [] **GitHub Personal Access Token** ([Generate](https://github.com/settings/tokens) (https://github.com/settings/tokens))
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5-Minute Setup

Step 1: Run Automated Setup

On Linux/Mac:

```
cd /home/ubuntu/codeforge_ai
chmod +x setup.sh
./setup.sh
```

On Windows:

```
cd C:\path\to\codeforge_ai
.\setup.ps1
```

Step 2: Configure API Credentials

Edit `backend/.env` with your credentials:

```
nano backend/.env
```

Add:

```
VENICE_API_KEY=your_venice_api_key_here
VENICE_API_SECRET=your_venice_api_secret_here
GITHUB_PERSONAL_ACCESS_TOKEN=ghp_your_token_here
GITHUB_USERNAME=your_github_username
```

Step 3: Start Backend

Terminal 1:

```
cd backend
source venv/bin/activate # On Windows: venv\Scripts\activate
python main.py
```

You should see:

```
🚀 CodeForge AI Backend Starting...
✅ Database connected
✅ Services initialized
INFO:      Uvicorn running on http://0.0.0.0:8000
```

Step 4: Start Frontend

Terminal 2:

```
cd frontend
npm run dev
```

You should see:

```
▶ Local:      http://localhost:3000
▶ Network:    http://192.168.x.x:3000
```

Step 5: Open Application

Open your browser and navigate to:

```
http://localhost:3000
```

First Tasks

1. Create Your First Project

1. Click **Projects** in the sidebar
2. Click the **+** button
3. Enter:
 - **Name:** "My First Project"
 - **Description:** "Testing CodeForge AI"
 - **Framework:** Select "React" or "Python"
4. Click **Create Project**

2. Generate Code with AI

1. Click **AI Chat** in the sidebar
2. Type a prompt:


```
Create a React component that displays a user profile with avatar, name, email, and a follow button
```
3. Watch the AI generate code in real-time!
4. Copy the code or save it to your project

3. Create an AI Agent

1. Click **Agents** in the sidebar
2. Click the **+** button
3. Configure:
 - **Name:** "Frontend Developer"
 - **Role:** "Frontend Development"
 - **Capabilities:** Add "React", "TypeScript", "CSS"
4. Click **Create Agent**

Your agent is now ready to handle tasks!

4. Push to GitHub

1. Select your project
2. Click the **GitHub** icon
3. Create a new repository:
 - **Name:** "my-first-codeforge-project"
 - **Description:** "Created with CodeForge AI"
 - **Private:** Choose Yes or No
4. Click **Create & Push**

Your code is now on GitHub!



Common Issues

Issue: "Database connection failed"

Solution:

```
# Check if PostgreSQL is running
sudo systemctl status postgresql

# Start PostgreSQL
sudo systemctl start postgresql

# Create database
sudo -u postgres psql -c "CREATE DATABASE codeforge_db;"
sudo -u postgres psql -c "CREATE USER codeforge WITH PASSWORD 'codeforge';"
sudo -u postgres psql -c "GRANT ALL PRIVILEGES ON DATABASE codeforge_db TO codeforge;"
```

Issue: "Port 8000 already in use"

Solution:

```
# Find and kill the process
lsof -ti:8000 | xargs kill -9

# Or change the port in backend/main.py
```

Issue: "Venice AI API error"

Solution:

1. Verify your API key in `backend/.env`

2. Check your API quota at [Venice.ai Dashboard](https://venice.ai/dashboard) (<https://venice.ai/dashboard>)

3. Test API connection:

```
bash
curl -X POST https://api.venice.ai/api/v1/chat/completions \
  -H "Authorization: Bearer YOUR_KEY" \
  -H "X-API-Secret: YOUR_SECRET" \
  -d '{"model": "dolphins-2.9.2-qwen2-72b", "messages": [{"role": "user", "content": "test"}]}'
```

Issue: “npm install fails”

Solution:

```
# Clear cache and reinstall
cd frontend
rm -rf node_modules package-lock.json
npm cache clean --force
npm install
```



Learn More

- **Full Documentation:** See [README.md](#) (README.md)
- **API Reference:** <http://localhost:8000/docs> (when backend is running)
- **Venice AI Docs:** <https://docs.venice.ai>
- **GitHub Integration:** See backend configuration



Pro Tips

1. **System Prompts:** Customize agent behavior by editing system prompts in the Agent creation form
2. **Code Editor Shortcuts:**
 - Ctrl+S / Cmd+S - Save file
 - Ctrl+F / Cmd+F - Find in file
 - Ctrl+/ / Cmd+/ - Toggle comment
3. **Agent Spawning:** Agents can automatically spawn child agents for specialized tasks. Watch the agent panel for spawning activity!
4. **Real-time Updates:** WebSocket connection provides live updates. Check the green indicator in the sidebar.
5. **Backup Your Work:**

```
```bash
Backup database
pg_dump -U codeforge codeforge_db > backup.sql
```

```
Export projects via GitHub integration
```

```
```
```

Next Steps

Now that you're set up, try:

1. **Generate a Full App:** Ask the AI to create a complete application
 2. **Multi-Agent Workflow:** Create multiple agents and assign them different parts of a project
 3. **GitHub Automation:** Set up automatic pushing of generated code
 4. **Custom System Prompts:** Experiment with different agent personalities and capabilities
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Important Notes

Localhost Notice: This application runs on your local machine:

- Backend: `http://localhost:8000`
- Frontend: `http://localhost:3000`

To access remotely, you'll need to deploy on a server or use port forwarding.

API Usage: Monitor your Venice AI API usage to avoid unexpected charges.

Security: Never commit `.env` files to version control!

✨ **You're ready to build with CodeForge AI!** ✨

Happy coding! 🚀